

ALEYSK, Russian Federation

WMO: 299370

Lat: 52.517N

Lon: 82.767E

Elev: 576

StdP: 14.39

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-29.7	-24.1	-35.4	0.7	-29.6	-29.8	1.1	-23.9	38.1	21.2	34.0	19.2	2.0	200	0.562

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.1	88.4	66.3	85.2	65.2	82.2	64.0	69.7	82.3	68.0	80.1	66.4	78.3	6.4	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.6	96.7	75.3	63.7	90.2	73.3	62.0	84.7	71.8	34.1	82.2	32.7	79.9	31.4	78.5	81.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation	n=5 years		n=10 years		n=20 years		n=50 years			
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(4) 31.1	22.9	18.6	DB	-32.9	94.2	7.1	4.0	-38.1	97.1	-42.2	99.5	-46.2	101.7	-51.4	104.7	
(5)			WB	-33.1	73.5	7.0	3.4	-38.2	76.0	-42.3	78.0	-46.3	79.9	-51.4	82.4	(p)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	38.0	2.2	7.1	22.1	42.3	55.8	66.4	68.6	64.6	53.0	40.3	22.3	9.0	(6)
	DBStd	25.80	15.01	13.76	11.82	10.31	9.17	7.01	5.81	6.62	7.79	9.25	14.19	13.80	(7)
	HDD50	6344	1482	1202	864	269	48	1	0	1	56	319	831	1271	(8)
	HDD65	10218	1947	1622	1329	680	306	69	31	89	364	765	1280	1736	(9)
	CDD50	1958	0	0	0	40	229	492	578	453	147	19	0	0	(10)
	CDD65	356	0	0	0	0	22	110	143	76	5	0	0	0	(11)
	CDH74	3768	0	0	0	26	372	1065	1264	892	144	5	0	0	(12)
	CDH80	1120	0	0	0	4	101	315	391	274	35	0	0	0	(13)
	WSAvg	6.1	6.7	6.4	6.6	6.8	6.2	5.1	4.1	4.2	4.8	6.6	7.6	8.2	(14)
	PrecAvg	18.0	1.0	0.8	1.0	1.1	1.6	2.1	2.7	1.7	1.3	1.5	1.7	1.4	(15)
(16) Precipitation	PrecMax	25.0	2.0	1.3	2.5	2.8	3.8	6.2	5.5	5.0	3.6	4.3	3.2	2.9	(16)
	PrecMin	11.1	0.2	0.0	0.1	0.1	0.5	0.6	0.6	0.2	0.2	0.4	0.6	0.4	(17)
	PrecStd	3.8	0.5	0.4	0.6	0.7	0.9	1.3	1.2	1.0	0.8	0.9	0.7	0.6	(18)
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.2	36.6	52.1	77.6	88.6	90.8	92.3	93.0	85.3	72.9	54.9	37.3	(19)
		MCWB	33.0	32.8	44.0	55.3	61.9	65.7	69.4	68.1	61.1	52.6	44.7	33.6	(20)
	2%	DB	31.6	32.9	44.9	70.3	82.6	87.0	88.0	86.6	78.6	65.3	48.4	33.9	(21)
		MCWB	29.6	30.2	38.9	52.6	59.5	64.7	68.4	66.2	59.0	50.2	41.7	31.5	(22)
	5%	DB	27.4	29.6	40.4	65.1	78.2	83.5	84.6	82.6	73.2	59.9	43.2	30.8	(23)
		MCWB	25.3	27.1	35.8	50.4	57.4	64.0	66.8	64.4	57.0	48.1	38.2	28.7	(24)
	10%	DB	23.3	26.1	37.0	59.3	73.5	80.3	81.3	79.0	68.3	54.4	38.5	27.7	(25)
		MCWB	21.5	24.0	33.6	47.4	56.1	63.5	65.9	63.2	54.5	45.6	35.2	25.8	(26)
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.5	33.8	44.3	57.2	65.1	70.0	73.3	70.5	64.0	54.1	46.5	34.1	(27)
		MCDB	35.8	36.0	51.4	73.2	83.2	82.6	86.2	85.5	79.5	66.4	51.3	36.4	(28)
	2%	WB	29.6	30.4	39.2	53.8	61.6	67.6	70.8	68.2	60.2	51.2	41.9	31.8	(29)
		MCDB	31.4	32.8	44.1	67.3	77.4	80.4	82.3	82.1	74.5	62.7	47.9	33.7	(30)
	5%	WB	25.4	27.2	36.2	51.1	59.5	66.1	68.9	66.2	58.0	48.9	38.5	28.8	(31)
		MCDB	27.2	29.5	40.0	63.0	73.6	79.2	80.5	79.4	70.8	59.0	42.9	30.6	(32)
	10%	WB	21.6	24.0	33.7	48.3	57.4	64.7	67.4	64.5	55.8	46.2	35.2	26.0	(33)
		MCDB	23.3	26.2	36.7	59.2	70.5	77.1	78.3	76.6	67.1	53.5	38.5	27.8	(34)
(36) Mean Daily Temperature Range	5% DB	MDBR	15.2	16.8	16.5	19.5	22.1	20.2	19.1	22.0	22.2	17.6	12.8	14.2	(35)
		MCDBR	16.2	17.4	17.9	29.8	29.3	25.8	24.4	28.5	31.7	29.1	16.4	16.6	(36)
		MCWBR	14.9	15.8	13.1	15.3	12.3	9.8	9.2	11.1	14.3	15.7	11.4	15.3	(37)
	5% WB	MCDBR	16.4	17.4	17.0	26.5	24.2	21.4	20.6	24.9	28.3	26.7	14.7	16.1	(38)
		MCWBR	15.2	16.1	12.9	14.9	11.7	9.2	9.2	10.7	13.6	15.3	11.0	15.3	(39)
(40) Clear-Sky Solar Irradiance	taub		0.301	0.338	0.391	0.419	0.418	0.413	0.437	0.412	0.374	0.361	0.347	0.310	(40)
	taud		2.014	1.944	1.898	2.096	2.168	2.264	2.197	2.241	2.290	2.210	2.039	1.962	(41)
	Ebn at Noon		203	231	243	257	264	267	257	257	253	226	181	169	(42)
	Edh at Noon		26	37	48	45	44	41	43	39	33	29	25	23	(43)
(44) All-Sky Solar Radiation	RadAvg		377	718	1175	1495	1830	2017	1820	1583	1149	613	323	263	(44)
	RadStd		29	45	118	105	155	135	118	94	99	76	41	34	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	-1.17	-0.94	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)	N/A	N/A	N/A	N/A	-0.95	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page